Quantitative Method Midterm

## WEEK 6 – PRINCIPLES OF GOOD DATA COLLECTION

### DATA COLLECTION

- Data collection is the process of gathering information needed to answer research questions or meet the objectives of a study.

### PRINCIPLES OF GOOD DATA COLLECTION

- Relevance – Collect only the data that is necessary to meet the objectives of the study. Avoid unnecessary or irrelevant data to save time and resources.
- Accuracy – Data should be correct, free from errors, and reflect reality.
- Use reliable tools and trained personnel to gather precise information.
- Consistency – Use standardized procedures, instruments, and definitions. Data should be collected in the same way across all participants and time periods.
- Validity – The data collected should truly measure what it is intended to measure. Ensure that the tools or questions used are appropriate for the target concept.
- Reliability – The method of data collection should produce stable and consistent results over time and across researchers.
- Timeliness – Data should be collected and available within a timeframe that allows it to be useful. Outdated data may not reflect current conditions or behavior.
- Confidentiality – Respect the privacy of participants. Data should be handled securely and only used for its intended purpose.
- Objectivity – Data should be collected impartially, without bias from the collector or respondent. Avoid influencing responses during interviews or surveys.
- Ethical Consideration – Obtain informed consent from participants. Avoid deception and protect vulnerable populations.
- Cost-effectiveness – Choose data collection methods that provide quality results without unnecessary expenses.

### GOOD DATA COLLECTION PRACTICES

- Good data collection practices ensure accuracy, reliability, validity, and ethics in data gathering.
- Important practices include collecting relevant data, using accurate and reliable tools, applying consistent procedures, protecting confidentiality, maintaining objectivity, obtaining informed consent, and using cost-effective methods.

### POOR DATA COLLECTION PRACTICES

- Poor data collection practices lead to inaccurate, biased, or unethical data.
- Important problems include collecting irrelevant information, using inaccurate instruments, inconsistent procedures, biased collection, failure to protect participant information, and failure to follow ethical considerations.

## WEEK 7 – DATA PRESENTATION AND DESCRIPTIVE STATISTICS

### DATA PRESENTATION

- Data presentation refers to the visual or tabular display of data to make it easier to understand.

### COMMON FORMS OF DATA PRESENTATION

Table – Presents data in rows and columns for organized viewing and comparison.

Bar Graph – Uses rectangular bars to compare quantities or categories.

Pie Chart – Uses sections of a circle to represent proportions or percentages of a whole.

Line Graph – Uses points connected by lines to show changes or trends, commonly over time.

Scatter Plot – Uses points to show the relationship or possible correlation between two variables.

### DESCRIPTIVE STATISTICS

- Descriptive Statistics is a branch of statistics that focuses on summarizing, organizing, and presenting data in an informative way.
- It is used to describe the basic features of data in a study and provides simple summaries about the sample and the measures.

### MAIN TYPES OF DESCRIPTIVE STATISTICS

- Measures of Central Tendency – Describe where the center of the data lies.

### MEAN

- The mean is the sum of all values divided by the number of values.

Formula: Mean = Sum of all values ÷ Number of values

Example: 12, 15, 10, 13, 20

Sum = 12 + 15 + 10 + 13 + 20 = 70

Number of values = 5

Mean = 70 ÷ 5 = 14

### MEDIAN

- The median is the middle value of a set of numbers when they are arranged in order.

Steps to find the median:

- Arrange the numbers from lowest to highest.
- If the number of values is odd, the median is the middle number.
- If the number of values is even, the median is the average of the two middle numbers.

### MODE

- The mode is the number that appears most frequently in a set of numbers.

Steps to find the mode:

- List all the numbers.
- Count how many times each number appears.
- The number that appears most often is the mode.

Example: 12, 15, 10, 13, 20, 12

12 appears the most.

Mode = 12

### MEASURES OF DISPERSION / VARIABILITY

- Measures of Dispersion describe how spread out the data is.

Range – The difference between the highest and lowest values.

Variance – The average of the squared differences from the mean.

Standard Deviation – The square root of the variance; shows how much values deviate from the mean.

### RANGE

Formula: Range = Highest Value − Lowest Value

Example: Scores = 8, 10, 6, 9, 7

Highest = 10

Lowest = 6

Range = 10 − 6 = 4
Variance and Standard Deviation Example:
Scores = 8, 10, 6, 9, 7
Sample Variance = 2.5
Sample Standard Deviation ≈ 1.58

## MEASURES OF POSITION

- Measures of Position describe the location or relative position of a value within a data set.

Percentiles – Values below which a certain percent of the data falls.
Examples: 25th percentile, 50th percentile, 75th percentile.
Quartiles – Values that divide a data set into four equal parts when the data is arranged in ascending order.

## FREQUENCY DISTRIBUTION

- A frequency distribution is a table that organizes data into categories or intervals and shows how often, or the frequency, each category or interval occurs.

Example: Test Scores of 20 Students
Raw scores: 45, 50, 55, 60, 65, 50, 55, 60, 45, 55, 65, 70, 60, 55, 50, 75, 70, 65, 55, 60
Interpretation:
Score 55 is the most common, with a highest frequency of 5.
Lowest score = 45.
Highest score = 75.
A frequency distribution provides a clear summary without listing all 20 scores.

# WEEK 8 AND 9 – PROBABILITY CONCEPT

## PROBABILITY

- Probability is the measure of how likely an event is to happen. It tells us the chance or possibility that something will occur.
- Probability is expressed as a number between 0 and 1, or 0% to 100%.

0 = The event is impossible.
1 = The event is certain.
Values between 0 and 1 show different levels of likelihood.

## BASIC PROBABILITY FORMULA

- Probability of an Event = Number of Favorable Outcomes ÷ Total Number of Possible Outcomes

Example – Rolling a Die:
Event E = Rolling a 4.
Total outcomes = {1, 2, 3, 4, 5, 6} = 6 outcomes.
Favorable outcome = {4} = 1 outcome.
P(E) = 1/6

Example – Tossing a Coin:
Event E = Getting Heads.
Total outcomes = {Head, Tail} = 2 outcomes.
Favorable outcome = {Head} = 1 outcome.
P(E) = 1/2

## BASIC PROBABILITY CONCEPTS

Experiment – A process that produces an outcome.
Examples: Rolling a die; tossing a coin.
Sample Space (S) – The set of all possible outcomes.
Example: Coin toss → S = {Heads, Tails}

Event (E) – A subset of the sample space.
Example: Getting a Head → E = {H}

Mutually Exclusive Events – Two events that cannot happen at the same time.
Example: Rolling a die and getting a 2 and getting a 5.

Independent Events – Two events where the outcome of one does not affect the other.
Example: Tossing two separate coins.

Complementary Events – Two events where if event A occurs, its complement A′ does not occur.

$P(A) + P(A') = 1$
$P(A)$ = Probability of event A occurring.
$P(A')$ = Probability of event A not occurring, or its complement.

COMPLEMENTARY EVENTS EXAMPLE
Coin Toss:
A = Getting a Head.
A′ = Getting a Tail.
$P(A) + P(A') = 1$.
Rolling a Fair Die:
A = Rolling an even number = {2, 4, 6}.
A′ = Rolling an odd number = {1, 3, 5}.

RULES OF PROBABILITY
- Addition Rule
- Multiplication Rule
- Conditional Probability

ADDITION RULE
- The Addition Rule is used when finding the probability of either Event A OR Event B happening.

Mutually Exclusive Events – Events cannot happen at the same time.
$P(A \text{ or } B) = P(A) + P(B)$
Example: Rolling a die and getting either a 2 or a 5. Since both cannot occur at the same time, their probabilities are added.

Not Mutually Exclusive Events – Events can happen at the same time, so the overlap must be subtracted.
$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Example: Drawing a card from a deck.

MULTIPLICATION RULE
- The Multiplication Rule is used to find the probability that two events happen at the same time or in sequence.

INDEPENDENT EVENTS
- If events A and B are independent, the occurrence of one does not affect the other.

$P(A \text{ and } B) = P(A) \times P(B)$
Example: Tossing a coin and rolling a die.

CONDITIONAL PROBABILITY
- Conditional Probability is the probability of an event B happening, given that another event A has already happened.

Formula: $P(B|A) = P(A \text{ and } B) \div P(A)$
$P(B|A)$ = Probability of B given A.
$P(A \text{ and } B)$ = Probability of both A and B happening.
$P(A)$ = Probability of A happening.
Example – Students:
Find P(glasses | left-handed).
40 students total.
10 are left-handed.
6 of the left-handed students wear glasses.
14 total students wear glasses.
P(glasses | left-handed) = 6/10 = 0.60 = 60%.
Meaning: If a student is left-handed, there is a 60% chance they wear glasses.

TYPES OF PROBABILITY
- Theoretical Probability
- Experimental Probability
- Subjective Probability
- Axiomatic Probability

THEORETICAL PROBABILITY
- Theoretical Probability is the likelihood of an event happening based on logic and known possible outcomes, without doing an actual experiment.
- It assumes that all outcomes are equally likely.

Formula: Theoretical Probability = Favorable Outcomes ÷ Total Possible Outcomes
Examples:

Coin Toss → Probability of getting Heads = 1/2.

Rolling a Die → Probability of getting a 3 = ⅙.

## EXPERIMENTAL PROBABILITY

- Experimental Probability is the likelihood of an event based on actual experiments, trials, or observations.
- It is calculated by performing an experiment and recording the outcomes.

Formula: Experimental Probability = Number of Times the Event Occurs ÷ Total Number of Trials

Examples:

- A survey of 100 students shows 25 like Math best.
- A coin is tossed 20 times and Heads appears 12 times.

## SUBJECTIVE PROBABILITY

- Subjective Probability is the likelihood of an event happening based on personal judgment, opinion, belief, or experience rather than exact calculation or experiments.
- It is often used when there is insufficient data or when the situation involves uncertainty.

Examples:

- A doctor estimates there is a 70% chance that a patient will recover based on medical experience.
- A student says, “I think there’s a 90% chance I’ll pass the exam,” based on confidence and preparation.
- A sports fan predicts their team has a 60% chance of winning based on recent performance.

## AXIOMATIC PROBABILITY

- Axiomatic Probability is a formal and mathematical approach to probability introduced by Andrey Kolmogorov in the 1930s.
- It is based on axioms, or rules, that probability must always follow regardless of the experiment.

## KOLMOGOROV’S THREE AXIOMS OF PROBABILITY

- Non-Negativity – The probability of any event is never negative.
- Normalization / Certainty – The probability of the sample space, containing all possible outcomes, is 1 or 100%.
- Additivity – If two events A and B are mutually exclusive, the probability of their union is the sum of their individual probabilities.

P(S) = 1

Example – Tossing a Coin:

Possible outcomes: Heads (H), Tails (T)

Sample Space: S = {H, T}

Because the coin is fair:

P(H) = 0.5

P(T) = 0.5

P(S) = P(H) + P(T) = 0.5 + 0.5 = 1

The total probability of all possible outcomes together must equal 1 or 100%. This means you are certain that when a fair coin is tossed, the outcome will be either Heads or Tails.

Example – Rolling a Die:

P(rolling an even number) = P(2) + P(4) + P(6)

= 1/6 + 1/6 + 1/6

= 3/6

= 0.5